

Charotar University of Science and Technology
M.Sc (Nano Science & Technology) Semester-II Examination May 2011

NT -708: Concepts of Physical and Chemical Sciences -II

Date: May 3, 2011

Time: 10.00 am to 10.30 am

Max. Marks:20

Part A

1. The solubility of o-phenol in water is less than that of m- and p- isomers. Explain. (2)
2. The dipole moment of phenol is lower than that of methanol. Justify. (2)
3. Calculate E for the half reaction $\text{Sn}^{+4} + 2e \rightarrow \text{Sn}^{+2}$ at 25°C , if the concentration of Sn^{+2} is four times that of Sn^{+4} . Given : $\text{Sn}^{+4} + 2e \rightarrow \text{Sn}^{+2}$ $E^\circ = 0.15\text{V}$ (3)
4. The rate constant of a chemical reaction is $2.3 \times 10^{-3} \text{ mol.lit}^{-1}.\text{sec}^{-1}$. The order of the reaction is:
i. First order, ii. Second order, iii. Zero order iv. Third order. (2)
5. How is the relative population of molecules at different energy levels related in Maxwell –Boltzmann distribution? (3)

6. Hardness of the metallic and ceramic materials: (2)
i. decreases with grain size . ii. decreases upto some critical size
iii. increases with grain size. iv. independent of grain size.
7. Optical properties change with: (2)
i. surface plasma. ii. surface area to volume ratio.
iii. number density of particles.
iv. both i and ii .
8. Crystalline material normally possesses ----- range ordering(long, short) (1)
9. When the particle size of Fe_3O_4 reduces to 10 nm from 500nm ,the domain structure will change from ----- to ----- (single domain, multi domain) (1)
- 10.For carbon dating the radioactivity of: (1)
i. C^{12} ii. C^{13} iii. C^{14} is measured.
- 11.For a first order reaction the concentration of the reactant at time follows : (1)
i. exponential decay,ii. linear decay not passing through origin .
iii. linear decay passing through origin,.

Charotar University of Science and Technology
M.Sc (Nano Science & Technology) Semester-II Examination May 2011

NT -708: Concepts of Physical and Chemical Sciences -II

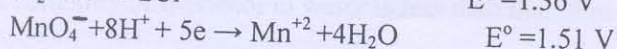
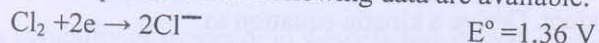
Date: May 3, 2011

Time: 10.30 am to 1.00 pm

Max. Marks: 50

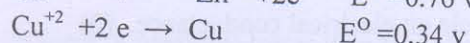
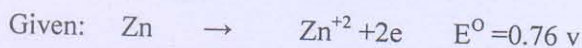
Part B

- 1(a). Is the oxidation of chloride ion to chlorine by the permanganate ion in an acid solution spontaneous? Following data are available: (5)



Calculate ΔG for the reaction.

- (b). Calculate the free energy change and equilibrium constant for the following reaction at 25°C : (5)



- 2(a). What is Nernst Heat Theorem? Explain the important consequence of this theorem? (4)

- (b). Consider the following data on a hypothetical reaction: (4)



Sr.No	[A] mol.dm ⁻³	[B] mol.dm ⁻³	Rate (r) mol.dm ⁻³ .sec ⁻¹
1.	6.0x10 ⁻³	1.0x10 ⁻³	0.012
2.	6.0x10 ⁻³	2.0x10 ⁻³	0.024
3.	2.0x10 ⁻³	1.5x10 ⁻³	0.002
4.	4.0x10 ⁻³	1.5x10 ⁻³	0.008

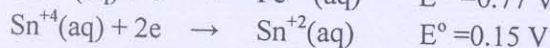
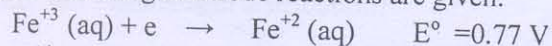
Answer the following:

- What is the order of the reaction with respect to A?
 - What is the order of the reaction with respect to B?
 - What is overall order?
- (c). The $T_{1/2}$ of a reaction is halved if the concentration of the reactant is doubled. Find the order of the reaction? (2)

- 3(a). Write in short about the origin of charge on colloidal particles. (2)

- (b). Why 2,6 -dihydroxy benzoic acid is much stronger than salicylic acid? (3)

- (c). The following electrode reactions are given: (3)



Show that reduction of Fe^{3+} is more spontaneous than that of Sn^{4+} .

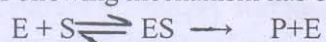
- (d). The rate of decomposition of a gas A was 7.25 in some unit, when 5% had reacted (2)

and it was 5.14 in the same unit when 20% had undergone decomposition. Find the order of the reaction.

- 4(a). Derive Langmuir's adsorption isotherm with proper reference to the basic assumptions involved. (5)

OR

- (a). Following mechanism has been proposed for enzyme catalyzed reactions:



Where E is enzyme, S is substrate and P is the product. Derive a kinetic equation to obtain the rate of formation of product.

- (b). Provide a suitable explanation for the relative stabilities of tertiary, secondary and primary carbocations. (5)

- 5(a). How does the magnetization of ferromagnetic materials vary with the size of the particles? Explain in short. (4)

- (b). Provide a brief account on the role of the size of particle on electrical conductance. (3)

- (c). In the nanometer size of grains, diffusion sliding greatly increases the deformation rate at room temperature. Is it true or false? Justify your answer. (3)